DATA SCIENCE & ANALYTICS · SESSION 05

Data Visualization & Storytelling

Choosing the right chart, building it well, and making it say something

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2026 · Bachelor Program · 3 hrs · Workshop Format · Hands-on Coding

Session Agenda

01 **Choosing the Right Chart**
Matching the chart to the question

02 **Matplotlib Fundamentals**
Figure, Axes, styling, and annotation

03 **Seaborn for Statistics**
lineplot, heatmap, pairplot, catplot

04 **Data Storytelling**
Situation → Complication → Resolution

05 **Annotating for Impact**
Making the chart make the point

06 **Workshop Block**
Tell a 3-chart story from your data — graded

01

PART 01

Choosing the Right Chart

The chart is a sentence about your data. Pick the type that answers the question you're actually asking — comparison, trend, relationship, or distribution.

Every chart type answers a different kind of question — match them

Chart selection	
Comparison	Bar charts compare a value across categories. The default when the question is 'which is biggest?'
Trend over time	Line charts show change across a continuous axis — months, days. Never use a line for unordered categories.
Relationship	Scatter plots show how two numbers relate. Add a third dimension with colour or bubble size.
Distribution	Histograms and boxplots show the spread of one variable. Heatmaps show a value across two dimensions.

Source: DS Course, Session 5, 2026

02

PART 02

Matplotlib Fundamentals

Matplotlib is the engine under every Python chart. Understand the Figure and Axes objects and you can build and control anything.

Figure and Axes: the two objects behind every Matplotlib chart

The mental model

Figure

The whole canvas — the window your charts live in. One Figure can hold several Axes in a grid.

Axes

A single plot: its own x/y axis, title, and data. plt.subplots() hands you both, and you draw onto the Axes.

Why care

The object API (ax.plot, ax.set_title) is explicit and scales to multi-panel figures, unlike scattered plt.* calls.

PYTHON · FIGURE & AXES

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(8, 4))
ax.plot(monthly.index, monthly.values)
ax.set_title('Monthly Revenue')
ax.set_xlabel('Month')
ax.set_ylabel('Revenue (THB)')
plt.tight_layout()
plt.show()
```

fig, ax = plt.subplots() is the line to memorize.

Two panels side by side: subplots for comparison

Small multiples

Grid of Axes

plt.subplots(1, 2) returns an array of Axes. Draw a different view in each to compare them honestly.

Share the scale

sharey=True keeps the y-axes identical so the comparison isn't misleading. Mismatched scales lie.

Tidy up

tight_layout() stops titles and labels from overlapping. A small habit that makes figures look professional.

PYTHON · SUBPLOTS

```
fig, axes = plt.subplots(1, 2,
    figsize=(12, 4), sharey=True)

avg.plot(kind='bar', ax=axes[0],
         color='steelblue')
axes[0].set_title('Average GPA')

counts.plot(kind='bar', ax=axes[1])
axes[1].set_title('Students')
```

Compare views in one figure with a shared scale.

03

PART 03

Seaborn for Statistics

Seaborn sits on top of Matplotlib and makes statistical charts — with confidence bands, heatmaps, and category breakdowns — one clean line each.

Seaborn: statistical charts in a single, readable line

Less code, more chart

High-level

Pass a DataFrame and column names; Seaborn handles grouping, colour, and the legend. Great for fast EDA visuals.

Statistical by default

lineplot draws a confidence band, boxplot shows quartiles — the statistics come built in, not bolted on.

Pairs with Matplotlib

It returns Axes, so you still fine-tune titles and labels with the Matplotlib calls you already know.

PYTHON · SEABORN

```
import seaborn as sns

# trend with a confidence band
sns.lineplot(data=df, x='month',
             y='revenue', hue='category')

# correlation heatmap
sns.heatmap(df.corr(numeric_only=True),
            annot=True, cmap='coolwarm')
```

Seaborn for the statistics, Matplotlib for the polish.

04

PART 04

Data Storytelling

A chart on its own is a fact. A story arranges facts so the audience reaches the conclusion with you — Situation, Complication, Resolution.

Three charts, three sentences: the SCR structure turns data into a decision

Situation → Complication → Resolution

Situation	Set the baseline everyone agrees on. 'Revenue grew steadily through 2023.' Chart one establishes the normal.
Complication	Introduce the tension. 'But Q4 margins fell as discounts rose.' Chart two reveals the problem worth caring about.
Resolution	Point to the action. 'Cut blanket discounts; protect margin in Electronics.' Chart three supports the recommendation.
Why it works	People remember a narrative, not a gallery of plots. SCR gives three charts a spine and a destination.

Source: DS Course, Session 5, 2026

05

PART 05

Annotating for Impact

The difference between a chart that decorates and one that persuades is the title, the highlight, and the one number you call out.

Make the chart make the point — don't leave the reader to find it

From label to insight

Title = the finding	'Revenue fell 18% after the March price change' beats 'Monthly Revenue'. State the conclusion in the title.
Highlight, don't list	Grey out the context and colour the one bar or line that matters. Direct the eye to your point.
Annotate the moment	An arrow and a few words on the key spike or drop save the reader from decoding the axes.
Less is more	Drop chartjunk — gridlines, 3-D, redundant legends. Every pixel that isn't the message is competing with it.

Source: DS Course, Session 5, 2026

06

PART 06

Workshop Block

Build three annotated charts that tell one coherent story from your dataset — Situation, Complication, Resolution. This is graded coding check #2.

By the end of this block, every student has a 3-chart story with a clear recommendation

Graded Check · 55 Minutes

Pick the question	Choose one decision your data can inform. The whole story serves that single question.
Build three charts	One each for Situation, Complication, Resolution. Right chart type for each, axes labelled.
Annotate	Title each chart with its finding. Highlight the one element that carries the point.
Write three sentences	One sentence per chart, ending in a recommendation. Charts plus sentences are your deliverable.

Source: Workshop, Session 5, 2026

KEY TAKEAWAY

A chart shows. A story convinces. Always build the story.